Lab 10

Weiwei Wang

2022-12-5

Deadline: 2022-12-11 22:30:00 Beijing Time

How to submit?

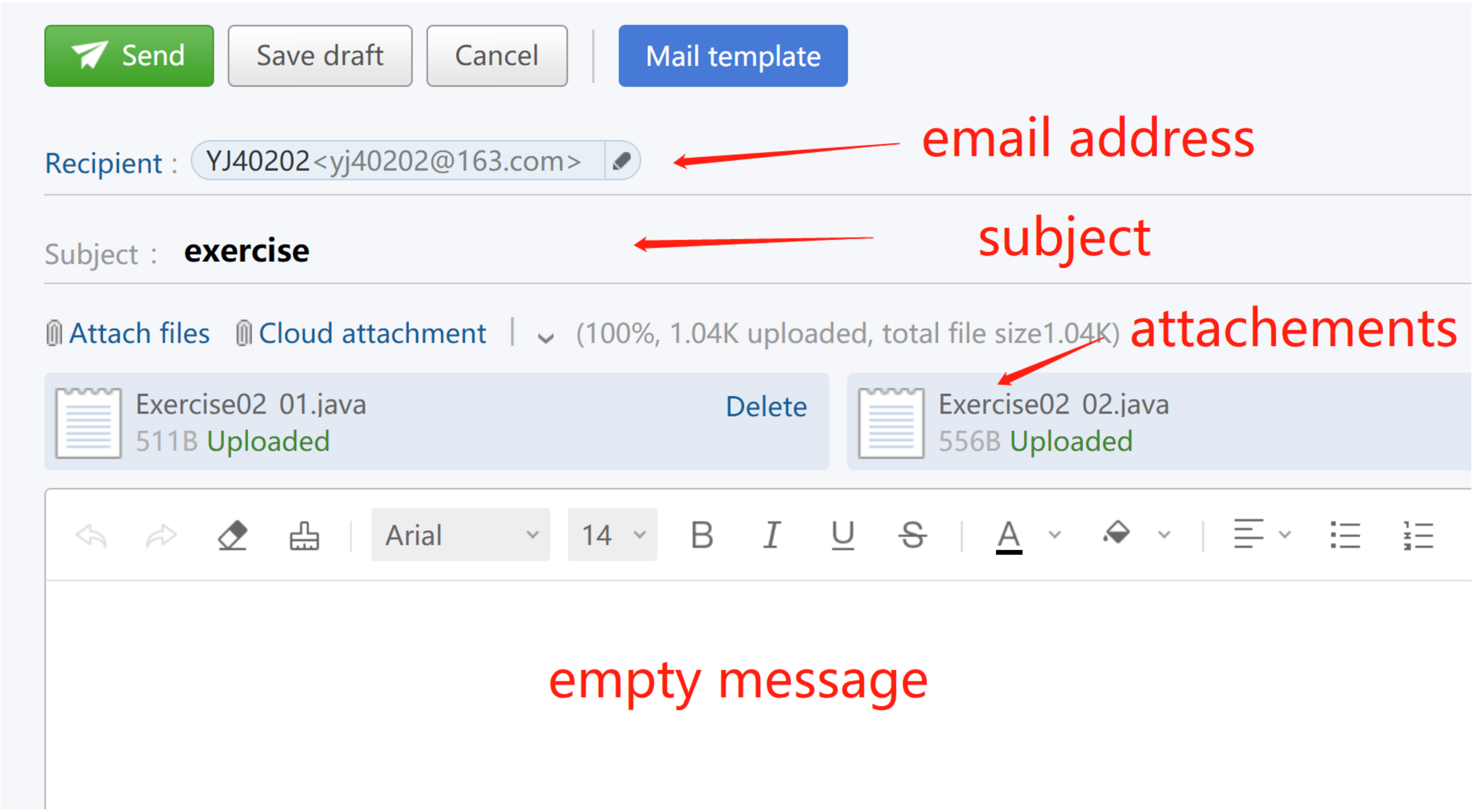
Email: YJ40202@163.com

Subject: exercise

Attachment: Your java source files.

Note: Don't use package.

How to submit?



Exercise11_01.java

(The Triangle2D class) Design a class named Triangle2D that extends GeometricObject. The class contains:

- Three double data fields named side1, side2, and side3 with default values 1.0 to denote three sides of a triangle.
- A no-arg constructor that creates a default triangle.
- A constructor that creates a triangle with the specified side1, side2, and side3.
- The accessor methods for all three data fields.
- A method named getArea() that returns the area of this triangle.
- A method named getPerimeter() that returns the perimeter of this triangle.
- A method named toString() that returns a string description for the triangle.

Exercise11_01.java

The toString() method is implemented as follows:

```
return "Triangle: side1 = " + side1 + " side2 = " + side2 +
 " side3 = " + side3;
```

Write a test program that prompts the user to enter three sides of the triangle and a color. The program should create a Triangle2D object with these sides and set the color using the input. The program should display the area, perimeter, and color. Here is a sample run

```
Enter three sides: 1 2 1.1
The area is 0.3197557617161977
The perimeter is 4.1
Triangle: side1 = 1.0 side2 = 2.0 side3 = 1.1
```

Exercise11_01.java

```java
class GeometricObject {
  private String color = "white";

  /** Construct a default geometric object */
  GeometricObject() {}

  /** Construct a geometric object with the specified color*/
  public GeometricObject(String color) {
         this.color = color;
  }

  /** Return color */
  public String getColor() {
    return color;
  }

  /** Set a new color */
  public void setColor(String color) {
    this.color = color;
  }
}
```

Exercise11_04.java

(Maximum element in ArrayList) Write the following method that returns the maximum value in an ArrayList of integers. The method returns null if the list is null or the list size is 0.

```
public static Integer max(ArrayList<Integer> list)
```

Write a test program that prompts the user to enter a sequence of numbers ending with 0 and invokes this method to return the largest number in the input.

Here is a sample run

```
Enter integers (input ends with 0): 1 3 5 -1 -10 -100 0
The maximum number is 5
```

Exercise11_10.java

(Implement MyStack using inheritance) Define a new stack class (MyStack) that extends ArrayList with the following methods:

- isEmpty(): boolean, Returns true if this stack is empty.
- getSize(): int, Returns the number of elements in this stack.
- peek(): Object, Returns the top element in this stack without removing it,
- pop(): Object, Returns and removes the top element in this stack.
- push(o: Object): void, Adds a new element to the top of this stack.

Write a test program that prompts the user to enter five strings and displays them in reverse order. Here is the sample run:

```
1 2 3 hello x [Enter]
x hello 3 2 1
```

Exercise11_13.java

(Remove duplicates) Write a method that removes the duplicate elements from an array list of integers using the following header:

```
public static void removeDuplicate(ArrayList<Integer> list)
```

Write a test program that prompts the user to enter 10 integers to a list and displays the distinct integers in their input order and separated by exactly one space.
Here is a sample run:

```
Enter ten integers:  34 5 3 5 6 4 33 2 2 4 [Enter]
The distinct integers are 34 5 3 6 4 33 2
```